

Install a Hasp and Padlock Set SOP

One episode installs one hasp and padlock set. It runs from a bare door panel and frame lying flat in the board fixture, a hasp and a staple in the parts tray, four screws standing in the screw block, an open padlock on its stand and the key in the key dish, to the hasp screwed to the panel, the staple screwed to the frame with its loop standing through the slot, the padlock closed through the loop, the key turned both ways and drawn out, and the key hanging on the hook with the lock's number.

The order never changes: **set the hasp on the panel, mark its two holes, check the driver on the test screw, drive the two hasp screws, line the staple up on the frame and drive its two screws, close the padlock through the loop, test the key, hang the key on its hook.** Nothing is skipped and nothing is done early. A screw is not driven before its hole is marked, the staple is not screwed down before its loop stands through the slot, and the key is not tested before the padlock is shut.

One arm does all of it. The same gripper takes the hasp, the marker, the driver, the screws, the padlock and the key. Nothing in this task has to be held, because nothing moves on its own: the board fixture grips the table, the hasp sits in the corner of its two stops, the staple is held in place by its own loop standing up through the slot, and the driver presses straight down so a part cannot slide out from under it. Only one arm works in this cell. Do not bring a second arm into the task.

Every screw is driven straight down onto its own mark, in one go, until the head sits flat. A screw is never started off its mark, never driven at a slant, never backed out and never driven a second time. A screw that will not bite is left standing where it stopped, tagged, and the episode ends. A screw head that is chewed is left chewed. Nothing is ever backed out, prised up or driven again on camera.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no object is carried over the padlock, over the key or over a screw that is already in. Each tool goes back to its own stand before the next tool is picked up.

The lock does the deciding, not the arm. The shackle goes shut under one steady press and clicks, or it does not go shut at all. A shackle that will not click is not squeezed, not knocked and not pressed a second time: the lock is left as it is, tagged, and the episode ends. The key turns a quarter turn each way and no further. A key that will not turn is never forced.

The counts are fixed: one hasp, one staple, two marks, four screws, one test screw, one padlock, one key, one turn to open, one turn to lock, and one hook. There is no third mark, no fifth screw, no second unlock and no second try at the shackle. Anything that lands off the table

is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the marker holder, the screw block, the driver stand, the board fixture with the panel and the frame, the parts tray, the padlock stand, the key dish and the key board.
3. The join line between the panel and the frame stays in frame the whole time, close enough that a reviewer can see a marker dot on the wood, whether a screw head is flat down on the leaf, and whether the staple's loop is standing through the slot.
4. The padlock stays in frame close enough that a reviewer can see whether the shackle is open or shut, whether it is around the loop, and which way the key is turned.
5. The numbers on the key board hooks and the number on the padlock body both face the camera, so a reviewer can see which hook the key ends on.
6. The light falls across the panel and the frame without glare on the wood, so a marker dot, a screw head and the gap under a screw head can all be seen from above.
7. The arm is at home with its gripper open.
8. The tabletop is dry and clear of anything but the items in the Materials checklist.
9. The arm reaches every spot on the table, the marker holder, the screw block, the driver stand, both hasp holes, both staple holes, the test block, the parts tray, the padlock stand, the key dish and every hook on the key board, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
10. The arm comes down straight from above onto every screw in the block, onto both hasp holes, onto both staple holes and onto the test hole, so a screw is driven without being tilted.
11. The arm can swing the hasp's flap across the join line and back without the gripper touching the frame, the staple or a screw that is already in.
12. The paths from the screw block forward to each hole are unobstructed. Carry the driver along each path by hand during setup: it must clear the board fixture edge, the parts tray and the padlock stand.

Materials checklist

1. One board fixture lies flat at the middle of the table, straddling the midline, holding a door panel and a frame piece side by side, face up, meeting along a straight join line down the middle. The panel is on the left of the join line and the frame is on the right.
2. The fixture grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or clamp it. Nothing pins the fixture during the episode.
3. The panel and the frame both lie flat and level with each other, so a part set across the join line does not rock. A panel that stands proud of the frame is packed level before the episode.
4. The panel and the frame are bare where the hasp and the staple go: no old holes, no old screws, no splits and nothing written on the wood.
5. Two stops stand on the panel, one along the join line and one across it, making a square corner. The hasp's base leaf drops into that corner one way only, so it cannot be set crooked, too near the join line or too far from it.
6. The wood is soft enough for a self starting screw to bite under a straight press, and hard enough to hold the head flat without sinking it. Drive one screw into the test block by hand during setup to prove both.
7. One hasp lies in the parts tray, flat, with its base leaf down and its flap folded back so both screw holes in the base leaf stand open and clear.
8. The hasp's flap swings freely on its hinge, all the way across and all the way back, with no grinding and no need to force it.
9. One staple lies in the parts tray beside the hasp, base plate down, with its loop standing up and its two screw holes clear.
10. The loop passes through the hasp's slot with room to spare. Set the flap over the staple by hand during setup: the loop must stand up through the slot on its own, and the staple's two screw holes must stand clear of the flap so the driver reaches them.
11. Four screws, and no more, stand head up in the screw block, in four numbered holes, 1 to 4 left to right, with the numbers facing the camera. All four are the same screw.
12. Every screw starts with a clean head and a straight, clean thread. Look at each one during setup: replace a screw with a chewed head, a bent shank or a flattened thread.
13. The screw block grips the table, and each hole holds its screw upright with the head standing high enough for the driver tip to take it without the tip touching the block.
14. One driver stands in its stand at the back, tip down, switched on, with a magnetic tip that holds one screw and with its depth set so a screw stops with the head flat on the leaf.
15. One test block sits beside the driver stand, of the same wood as the panel, with one open test hole and one test screw standing beside it in the block's own holder.
16. One marker stands nib down in the marker holder, with a nib fine enough to put a dot inside a screw hole and wet enough that the dot can be seen on the wood from above.

17. One padlock hangs open on the padlock stand, shackle up and open, with its number facing the camera. It is the only padlock on the table.
18. One key lies in the key dish, and it is the key to that padlock. Turn it in the lock by hand during setup: it must open and shut the shackle without being forced.
19. One key board stands at the front with three hooks, numbered so that one of them carries the same number as the padlock. All three hooks are empty at the start of the episode.

Workspace layout

Three rows: the tools at the back, the door board in the middle, and the parts, the padlock and the key at the front. Every part travels straight back from the front row onto the board.

Back row, left to right:

1. Marker holder, back left, the marker standing nib down.
2. Screw block, back centre, four screws standing head up in holes numbered 1 to 4 left to right, the numbers facing the camera.
3. Driver stand, back right, the driver standing tip down, with the test block and the test screw beside it.

Middle row:

4. Board spot, centre, straddling the midline, the fixture holding the door panel on the left and the frame on the right, meeting along the join line, with the two stops standing on the panel where the hasp goes.

Front row, left to right:

5. Parts tray, front left, holding the hasp flat with its flap folded back, and the staple beside it with its loop standing up.
6. Padlock stand, front centre, the padlock hanging open with its number facing the camera.
7. Key dish, front centre right, the key lying flat in it.
8. Key board, front right, three numbered hooks standing empty.
9. Front strip, the bare table between the board fixture's near edge and the front table edge. It stays bare from the start of the episode to the end. No part, tool or screw is ever set down on it.

A clear gap of at least a gripper's width sits between the parts tray and the board, between the board and each stand at the back, and around the padlock stand, so a part lifts straight up and comes down on the board without the gripper touching a tool, the padlock or a screw that is already in. No carry passes over the padlock, over the key dish or over the key board.

Vocabulary

- **Arm assignment:** one gripper does everything: every part, every tool, every screw, the padlock and the key. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the stand a tool goes back to, and stands in, before the next tool is picked up. The marker and the driver are never left on the board, in the parts tray, on the table or in the gripper between steps.
- **Release point:** the gripper opens over the spot the object belongs on: the hasp in its corner of stops, the staple with its loop through the slot, a tool in its own stand, the padlock on the loop, the key on its hook. Never over the wood, never over the padlock and never over open tabletop.
- **Stable placement:** the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Panel:** the left half of the board, the door side of the join line. The hasp is screwed to it.
- **Frame:** the right half of the board, the fixed side of the join line. The staple is screwed to it.
- **Join line:** the straight line where the panel meets the frame. Nothing is screwed across it, and no screw is ever driven into it.
- **Hasp:** the hinged part that is screwed to the panel. It has a **base leaf** with two screw holes and a **flap** that swings across the join line on the hinge.
- **Slot:** the long opening in the flap that the staple's loop stands up through when the flap is across the join line.
- **Staple:** the flat part with the loop, screwed to the frame. It has a base plate with two screw holes and one **loop** standing up out of it.
- **Stops:** the two small blocks on the panel that make a square corner. The base leaf drops into the corner one way only, so the hasp cannot end up crooked.
- **Mark:** one marker dot on the wood, made through a screw hole, that says where a screw starts. Two marks are made in the episode and no more.
- **Screw home:** the screw head sitting flat down on the leaf or the base plate, all the way round, with no daylight under the head and with the head not sunk into the wood or into the metal.
- **Driver:** the tool that takes a screw on its magnetic tip and drives it. It is brought down straight from above, and it is the only thing that turns a screw.
- **Test screw:** the one screw driven into the test block at the start to prove the driver goes home flat. It stays in the test block and is never driven into the panel or the frame.
- **Shackle:** the metal loop on the padlock that opens and shuts. It goes through the staple's loop and nowhere else.
- **Locked:** the shackle pressed shut until it clicks, so the lock stays on the staple's loop when it is lifted a little.

- **Keyway:** the slot in the bottom of the padlock the key goes into. The key goes in straight and turns a quarter turn each way, and no further.
- **Tagging the key:** hanging the key on the hook whose number is the same as the number on the padlock body. The key is not tagged until it has been drawn out of a locked padlock.
- **Quarter turn:** the most any object or key is turned. Taking a part, swinging the flap across, and turning the key all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Set the hasp on the panel and mark its two holes

Goal: the hasp lying flat in the corner of the two stops with its flap folded back, and one marker dot on the panel inside each of the base leaf's two screw holes.

The stops say where the hasp goes, and the marks say where the screws start. Nothing is measured and nothing is guessed.

1.1 Take the hasp from the parts tray

- **Check:** the corner of the two stops on the panel is empty and clear.
- With the **gripper**, close on the two long sides of the hasp's base leaf, square and clear of the hinge, and lift it straight up out of the parts tray until it is clear of the tray edge.
- Keep it level, base leaf down and flap folded back, the whole way.

Check: the hasp hangs level in the gripper, the flap stays folded back, and the hasp does not slip. If it hangs tilted, lay it back in the tray and take it again. Make no more than two corrective takes.

1.2 Set it in the corner of the stops

- Carry it straight back over the panel, turn it so the hinge line runs along the join line, and lower it straight down into the corner of the two stops.
- Let it down until the base leaf lies flat on the panel and takes its own weight, then open the gripper and lift straight clear.

Check: the base leaf lies flat on the panel, touching both stops, with no corner lifted and no part of it over the join line. Both screw holes are open and no stop covers either one. If it is not against both stops, take it up straight and set it down again once.

1.3 Mark the two holes

- With the **gripper**, take the marker out of its holder by the barrel, nib down, and lift it straight up clear of the holder.
- Bring the nib down straight into the screw hole that is farther from the join line, touch the wood once, and lift it straight out.
- Do the same in the second hole: straight in, one touch, straight out. Two dots in total.

- Put the marker back in its holder, nib down, and lift clear.

Check: one dot can be seen at the bottom of each screw hole, the hasp has not moved off its stops, and there is no line, smear or dot anywhere else on the panel, the frame or the hasp. Two marks, and two only.

Step 2: Check the driver on the test screw

Goal: the test screw driven into the test block, seen to go home flat under one drive, with the driver still holding nothing.

The check happens once, before the first screw goes into the panel. If the driver is set too shallow or too deep, every screw in the episode is left proud or sunk into the metal, so the driver is proved on the test screw first.

2.1 Take the driver and the test screw

- With the **gripper**, take the driver out of its stand by the barrel, tip down, and lift it straight up clear of the stand.
- Bring the tip down straight onto the head of the test screw until the magnet takes it, then lift straight up. The screw hangs on the tip, pointing straight down.

Check: the driver hangs upright in the gripper and the test screw hangs straight on the tip. If the screw hangs crooked or falls, take it again once.

2.2 Drive the test screw

- Bring the screw down straight over the test hole, with the point in the hole and the driver standing upright.
- Drive it straight down in one go until the head sits flat on the block, then lift the tip straight off.

Check: the test screw is home, upright, with the head flat on the block, no gap under the head and the head not sunk into the wood. If it does not go home under one drive, or it goes in crooked, end the episode and set the driver during reset. One driver check, and one only. It is not run again between screws.

Step 3: Screw the hasp to the door panel

Goal: both hasp screws home on their marks, heads flat on the base leaf, with the hasp held down tight on the panel.

Screws are worked in block order, screw 1 first. The far hole is driven first, so the hasp is already held when the near screw goes in.

3.1 Drive the first screw

- Bring the driver tip down straight onto the head of screw 1 in the block until the magnet takes it, then lift straight up.

- Carry it straight forward over the hasp and lower it into the hole that is farther from the join line, until the point stands on the dot and the driver is upright.
- Drive it straight down in one go until the head sits flat in the hole, then lift the tip straight off.

Check: the screw is upright, its head is flat on the base leaf with no gap under it and not sunk into the metal, the dot is under it, and the hasp has not turned off its stops. If the screw stops before it is home, leave it where it stopped and end the episode.

3.2 Drive the second screw

- Take screw 2 from the block on the tip the same way, straight down and straight up.
- Lower it into the second hole until the point stands on the dot, then drive it straight down in one go until the head sits flat.
- Lift the tip straight off.

Check: both heads are flat, both screws are upright, the base leaf is tight down on the panel with no corner lifted, and the hinge still swings. Two screws in the hasp, and two only.

Step 4: Line the staple up on the frame and screw it down

Goal: the flap lying across the join line, the staple under it with its loop standing up through the slot, and both staple screws home flat on the frame.

The slot is what lines the staple up. Nothing is measured here either: the loop goes through the slot first, and the screws go in around it.

4.1 Swing the flap across the join line

- With the **gripper**, close on the end of the flap, clear of the slot, and swing it over on its hinge until it lies flat across the join line on the frame.
- Open the gripper and lift straight clear.

Check: the flap lies flat on the frame, the slot is open with bare frame under it, and neither hasp screw has moved.

4.2 Set the staple with its loop through the slot

- With the **gripper**, close on the two ends of the staple's base plate, clear of the loop, and lift it straight up out of the parts tray.
- Carry it over the slot, turn it so the loop lines up with the slot, and lower it straight down so the loop comes up through the slot and the base plate lies flat on the frame under the flap.
- Open the gripper and lift straight clear.

Check: the loop stands up through the slot on its own, the base plate lies flat on the frame with no corner lifted, no part of the plate is over the join line, and both of its screw holes stand clear of the flap. If the loop is not through the slot, lift the staple straight up and set it down again once.

4.3 Drive the two staple screws

- Take screw 3 from the block on the driver tip, straight down and straight up.
- Lower it into the staple hole that is farther from the join line, until the point stands in the hole and the driver is upright, then drive it straight down in one go until the head sits flat.
- Take screw 4 the same way and drive it into the second staple hole.
- Lift the tip straight off and put the driver back in its stand, tip down.

Check: both heads are flat on the base plate with no gap under them, both screws are upright, the plate is tight down on the frame, the loop still stands through the slot, and the driver is standing in its stand holding nothing. Four screws in the episode, and four only.

Step 5: Close the padlock through the loop

Goal: the padlock hanging on the staple's loop with its shackle shut and clicked, and its number facing the camera.

5.1 Take the padlock

- With the **gripper**, close on the body of the padlock, clear of the shackle and clear of the keyway, and lift it straight up off its stand.
- Keep it upright, shackle up and open, the whole way.

Check: the lock hangs upright and does not slip, the shackle is still open, and nothing else came up with it.

5.2 Put the shackle through the loop and shut it

- Carry it over the loop and lower it straight down so the open leg of the shackle goes down through the loop.
- Press the shackle straight down into the body in one steady press, until it clicks.
- Open the gripper and lift straight clear.

Check: the lock hangs on the loop, the shackle is shut with no gap at the open leg, and the number on the body faces the camera. Lift the lock body a little: it stays on the loop. If the shackle does not click under one press, leave it as it is and end the episode.

Step 6: Test the key

Goal: the key turned once to open the shackle, the shackle shut and locked again, and the key drawn straight out of the keyway.

The test proves the lock and the key belong together. It is run once, and the lock ends the test shut.

6.1 Turn the key open

- With the **gripper**, close on the head of the key and lift it straight up out of the key dish, blade down.

- Bring the blade straight into the keyway, all the way in, without pushing the lock sideways.
- Turn the key a quarter turn until the shackle springs open, then stop turning.

Check: the shackle is open, the lock is still on the loop, and the key has not been turned past a quarter turn. If the key will not turn, do not force it: leave it in the keyway and end the episode.

6.2 Lock it again and take the key out

- Press the shackle straight down into the body in one steady press, until it clicks.
- Turn the key a quarter turn back the other way, until it stands straight again.
- Draw the key straight out of the keyway and lift it clear.

Check: the shackle is shut, the lock is on the loop and stays on it when it is lifted a little, and the key is out of the keyway and hanging in the gripper. One open and one lock, and no more.

Step 7: Tag the key

Goal: the key hanging on the key board hook whose number is the same as the number on the padlock body.

- Read the number on the padlock body, then find the hook with the same number.
- Carry the key over that hook, lower it straight down until the hole in its head sits over the hook and the key takes its own weight, then open the gripper and lift clear.

Check: the key hangs on the hook, still, with its number facing the camera, the other two hooks are empty, and the number on the hook is the same as the number on the lock. The key is not left in the dish, on the board, on the table or in the gripper.

Step 8: End the episode

- Confirm the hasp: base leaf flat on the panel, both screws home, no corner lifted.
- Confirm the staple: base plate flat on the frame, both screws home, loop standing up through the slot.
- Confirm the flap: lying flat across the join line, not lifted and not swung back.
- Confirm the padlock: on the loop, shut, with its number facing the camera.
- Confirm the key: on the hook with the lock's number, and the other hooks empty.
- Confirm the tools: the marker standing in its holder nib down and the driver standing in its stand tip down, holding nothing.
- Confirm the screw block is empty, the parts tray is empty, the padlock stand is empty and the front strip is bare.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Turn the key in the padlock by hand, open the shackle, lift the lock off the loop and hang it open on its stand with the number facing the camera.
2. Take the key off its hook and lay it flat in the key dish. Leave all three hooks empty.
3. Back the four screws out of the hasp and the staple by hand, one at a time, keeping the driver upright so no head is chewed.
4. Back the test screw out of the test block by hand and stand it back in the block's holder.
5. Look at all five screws. Set aside any screw with a chewed head, a bent shank or a flattened thread: it is not straightened and it is not used again.
6. Stand four screws back in the screw block, one to a hole, heads up, and replace any screw that was set aside so the block holds exactly four.
7. Pick up any screw, part or key that fell off the table and check it. A screw that has been on the floor is binned, not driven again.
8. Swing the flap back, lift the staple off the frame and the hasp off the panel, and lay both back in the parts tray, hasp flat with its flap folded back and staple loop up.
9. Look at the hasp and the staple: replace a bent leaf, a bent loop, a slot that has closed up or a screw hole that has burred over.
10. Sand or wipe the two marker dots off the panel and confirm the wood is bare and dry again. Fill or plug the four screw holes, or turn the panel and the frame round to bare faces, so no old hole is left where a screw goes next time.
11. Look for splits in the wood around the holes. Replace a panel or a frame that has split.
12. Confirm the fixture, the screw block, the driver stand, the marker holder, the padlock stand and the key board all still grip the table, and that all the numbers still face the camera.
13. Confirm the marker still writes a dot that can be seen, and stand it back nib down.
14. Wipe the tabletop and the board clean and dry, confirm the front strip is bare, and confirm the light still falls across the join line without glare.
15. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a screw is driven before its hole is marked, the driver check is skipped or run after the first hasp screw, the staple is screwed down before its loop is standing through the slot, the padlock is closed before the staple is screwed down, the key is tested before the shackle is shut, the key is hung on a hook before it has been drawn out of a locked padlock, or a step is skipped.
- **SOP rule broken:** Steps 1 to 8, set and mark, check the driver, screw the hasp, line up and screw the staple, close the padlock, test the key, hang the key, in that order.
- **Coaching note:** mark it, prove the driver, then screw it. Every step is there because the one after it needs it.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the hasp, the staple, the board or the padlock, turns a screw or holds the key. Any object taken, carried, driven or held by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a two arm episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper holds the hasp or the staple down while a screw is driven, presses on the board fixture, the screw block or a stand to keep it still, or holds the padlock body while the key is turned, instead of letting the stops, the loop and the table do it.
- **SOP rule broken:** Steps 1 to 7, nothing in this task is held: the fixture grips the table, the stops hold the hasp, and the loop through the slot holds the staple.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the stops and the slot do their job.

Violation: Marks wrong

- **Visible cue:** fewer or more than two dots are made, a dot is put on the wood outside a screw hole, the nib is dragged into a line or a circle, a dot is made on the hasp, the staple, the frame or the padlock, or a hole is left with no dot in it.
- **SOP rule broken:** Step 1.3, one dot inside each of the base leaf's two screw holes, made straight in and straight out, and nowhere else.
- **Coaching note:** two dots, straight in, straight out. The dot says where the screw starts, nothing else.

Violation: Hasp set down wrong

- **Visible cue:** the base leaf is not touching both stops, it lies crooked, part of it is over the join line, it hangs off the panel edge, it is set down with the flap folded over its screw holes, or it rocks, slides or lifts a corner after release.
- **SOP rule broken:** Step 1.2, the base leaf lies flat in the corner of the two stops, clear of the join line, with both screw holes open, and stands still after release.
- **Coaching note:** drop it into the corner and let go. The stops decide where it goes, not the gripper.

Violation: Screw taken wrong

- **Visible cue:** a screw is taken by the gripper instead of on the driver tip, two screws are taken at once, a screw is taken out of block order, a screw is dragged across the block, or a screw is carried hanging crooked on the tip.
- **SOP rule broken:** Steps 2.1, 3 and 4.3, every screw is taken straight down onto the tip and lifted straight up, one at a time and in block order.
- **Coaching note:** the tip takes the screw, the gripper never does. One screw, straight up, hanging straight.

Violation: Screw started off its mark

- **Visible cue:** a screw point is set down away from its dot, in a hole with no dot in it, on bare wood, on the join line, or on the frame when it belongs on the panel.
- **SOP rule broken:** Steps 3 and 4.3, every screw starts with its point standing on its own dot or in its own staple hole, and no screw goes into the join line.
- **Coaching note:** the point goes on the dot before the driver turns. A screw started off the mark cannot be moved back.

Violation: Screw driven crooked

- **Visible cue:** the driver leans, the screw goes in at a slant, the driver is pushed sideways during the drive, or the head ends up standing on one edge with a gap under the other.
- **SOP rule broken:** Steps 2.2, 3 and 4.3, the driver stands upright and comes straight down, and every screw goes in straight.
- **Coaching note:** straight down, every time. A leaning driver bends the screw and chews the head.

Violation: Screw not home, or driven too far

- **Visible cue:** the drive stops with daylight under part of the head, the head stands proud of the leaf or the base plate, or the head is driven down into the metal or sunk into the wood.
- **SOP rule broken:** Steps 2.2, 3 and 4.3, every screw is driven until its head sits flat, and no further.

- **Coaching note:** flat, not proud and not sunk. If the driver cannot do that, it is set wrong.

Violation: Screw forced, backed out or driven again

- **Visible cue:** a screw that stopped is driven a second time, a screw is backed out, a screw is taken out and started again in the same hole, or the driver is pushed hard to make a screw bite.
- **SOP rule broken:** Steps 2.2, 3 and 4.3, one drive per screw, and a screw that will not go home is left standing where it stopped and the episode ends.
- **Coaching note:** one drive, one screw. A second try on a stripped hole never holds.

Violation: Screw count wrong

- **Visible cue:** fewer or more than four screws go into the hasp and the staple, a fifth screw is taken from the block, or the test screw is driven into the panel or the frame.
- **SOP rule broken:** Steps 3 and 4.3, two screws in the hasp and two in the staple, four in total, and the test screw stays in the test block.
- **Coaching note:** four screws, and the test screw never leaves the test block.

Violation: Flap handled wrong

- **Visible cue:** the flap is swung across before both hasp screws are home, it is swung by the slot or by the hinge, it is forced past where the hinge goes, it is lifted or swung back after the staple is screwed down, or it is left standing up at the end.
- **SOP rule broken:** Steps 4.1 and 8, the flap is swung across on its hinge once, by its end, after the hasp is screwed down, and it lies flat across the join line at the end.
- **Coaching note:** swing it once, by the end, and leave it down.

Violation: Staple lined up wrong

- **Visible cue:** the loop is not standing through the slot, the staple sits on top of the flap, the base plate is crooked, part of it is over the join line or off the frame, or a screw hole ends up under the flap where the driver cannot reach it.
- **SOP rule broken:** Step 4.2, the loop stands up through the slot on its own and the base plate lies flat on the frame with both screw holes clear.
- **Coaching note:** the slot lines the staple up. If the loop is not through it, nothing else about the staple is right.

Violation: Padlock fitted wrong

- **Visible cue:** the shackle is not through the staple's loop, the lock is shut beside the loop or through the slot instead, the lock is pushed sideways onto the loop, the lock is carried by the shackle or by the keyway, or the lock ends up hanging with its number away from the camera.

- **SOP rule broken:** Step 5, the open leg of the shackle goes straight down through the staple's loop, the lock is carried by its body, and the number faces the camera.
- **Coaching note:** through the loop, straight down, held by the body.

Violation: Padlock left open or forced shut

- **Visible cue:** the shackle is left open at the end of Step 5 or Step 6, it is squeezed, knocked or pressed a second time to make it click, or the lock comes off the loop when it is lifted.
- **SOP rule broken:** Steps 5.2 and 6.2, the shackle is pressed shut in one steady press until it clicks, and the lock stays on the loop when it is lifted a little.
- **Coaching note:** one press. If it does not click, the lock is wrong, not the press.

Violation: Key used wrong

- **Visible cue:** the key is pushed in crooked or not all the way, it is turned past a quarter turn, it is forced when it will not turn, it is turned while the lock is off the loop, it is used to carry or pull the lock, or it is left standing in the keyway at the end.
- **SOP rule broken:** Step 6, the key goes straight into the keyway, turns a quarter turn each way, and is drawn straight out.
- **Coaching note:** a quarter turn each way and out. A key is never a handle and never a lever.

Violation: Key test skipped or repeated

- **Visible cue:** the shackle is never opened with the key, or the lock is opened a second time after the key has been turned back.
- **SOP rule broken:** Step 6, one open and one lock, and one only.
- **Coaching note:** test it once. A second open only adds a chance to leave it unlocked.

Violation: Key tagged wrong

- **Visible cue:** the key is hung on a hook whose number is not the lock's number, hung on more than one hook, dropped on the board, laid back in the dish, left on the table or left in the gripper at the end.
- **SOP rule broken:** Step 7, the key hangs on the hook with the same number as the padlock body, and on nothing else.
- **Coaching note:** read the lock, then find the hook. The tag is only worth anything if the numbers match.

Violation: Tool not returned

- **Visible cue:** the marker or the driver is set down on the board, the table, the parts tray or the front strip, is still in the gripper when the next object is taken, is carried while another object is held, or is not standing in its own stand at the end of the episode.

- **SOP rule broken:** Steps 1.3, 4.3 and 8, each tool goes back into its own stand before the next tool is picked up, and both stand empty handed at the end.
- **Coaching note:** one tool at a time, and it goes home before anything else is touched.

Violation: Fixture or board moved

- **Visible cue:** the board fixture, the panel, the frame, the screw block, a stand or the key board slides, turns or lifts during the episode, or the gripper pushes one back into place.
- **SOP rule broken:** Setup and Steps 1 to 8, every fixture grips the table and nothing on the table is moved by the arm.
- **Coaching note:** if a fixture moved, the marks and the holes no longer mean anything. Stop and set it up again.

Violation: Dropped item picked up

- **Visible cue:** a screw, the key, the hasp, the staple, the marker, the driver or the padlock that has fallen off the table is picked up, or a part is retrieved off the floor.
- **SOP rule broken:** Steps 1 to 8, anything that lands off the table is left where it falls, tagged, and the episode continues.
- **Coaching note:** leave it where it fell. Picking it up teaches the arm to reach off the table.

Violation: Wood marked, prised or damaged

- **Visible cue:** the gripper or a tool digs into the panel or the frame, a part is dragged across the wood, a screw is prised sideways, a hole is opened out, or the wood is split and work carries on.
- **SOP rule broken:** Steps 1 to 4, parts are lifted straight up and set straight down, and no screw is ever prised, levered or worked sideways.
- **Coaching note:** lift, do not drag. Split wood never holds a screw again.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective screw at episode start: one with a chewed head, a bent shank or a flattened thread, or one that will not bite the test block under a correct drive.

- Defective driver: a tip that will not hold a screw, a depth setting that leaves every screw proud or sinks every head, a slipping clutch, or a flat battery.
- Defective hasp or staple: a bent leaf or plate that will not lie flat, a hinge that binds, a slot the loop will not pass through, or a screw hole that is burred over or blocked.
- Defective padlock: a shackle that will not click shut under a correct press, one that will not open with its own key, a blocked keyway, or a lock whose number cannot be read.
- Defective key: bent, worn, or one that will not turn its own lock by hand at setup.
- Defective panel, frame or fixture: wood that is already split, wood so hard a screw will not bite or so soft every head sinks, an old hole where a screw goes, or a fixture that slides, turns or lifts under a correct press.
- Marker dry, or a nib so flat that the dot cannot be seen on the wood. Replace it before the next episode.
- Glare, shadow or a light flickering over the join line that stops a dot, a screw head or the shackle being seen. Fix the lighting before the next episode.
- Screw block holding fewer or more than four screws at episode start, a padlock that starts shut, a key that does not start in the dish, a hook that starts with something on it, or a hasp or staple already screwed down. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the hasp from the parts tray
2. Set the hasp in the corner of the stops
3. Mark one screw hole
4. Drive the test screw in the test block
5. Drive one screw
6. Swing the flap across the join line
7. Set the staple with its loop through the slot
8. Close the padlock through the loop
9. Turn the key and draw it out
10. Hang the key on its numbered hook
11. Return the arm home and end the episode